

● EASY

● ALGEBRA

● ANSWER KEY

Algebra

10 answers · Rationale-rich · Teach from doc

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Q1**D****D.** 902

Choice D is correct. Substituting 9 for x in the given equation yields $f9 = 1009 + 2$, or $f9 = 902$. Therefore, the value of fx when $x = 9$ is 902.

Choice A is incorrect. This is the value of fx when $x = 1.09$.

Choice B is incorrect. This is the value of fx when $x = 1.16$.

Choice C is incorrect. This is the value of fx when $x = 8.98$.

Q2**B****B.** 32

Choice B is correct. Dividing both sides of the given equation $7x = 28$ by 7 yields $x = 4$. Substituting 4 for x in the expression $8x$ yields 32, which is equivalent to 32.

Choice A is incorrect. This is the value of $\frac{21}{4}x$.

Choice C is incorrect. This is the value of $42x$.

Choice D is incorrect. This is the value of $56x$.

Q3**B****B. 32**

Choice B is correct. Since Brittany's total take-home pay was \$1,576, the value 1,576 can be substituted for T in the given equation $T = 1,000 + 18h$ to give $1,576 = 1,000 + 18h$. Subtracting 1,000 from both sides of this equation gives $576 = 18h$. Dividing both sides of this equation by 18 gives $32 = h$. Therefore, Brittany was paid for 32 hours for her first week of work.

Choice A is incorrect. This is half the number of hours Brittany was paid for. Choice C is incorrect and may result from dividing 1,000 by 18. Choice D is incorrect and may result from dividing 1,576 by 18.

Q4**55**

The correct answer is 55. Subtracting 40 from both sides of the given equation yields $x = 55$. Therefore, the value of x is 55.

Q5**1**

The correct answer is 1. It's given that the function f is defined by $fx = x + \frac{8}{11}$. Substituting $\frac{3}{11}$ for x in the given function yields $f\frac{3}{11} = \frac{3}{11} + \frac{8}{11}$, which gives $f\frac{3}{11} = \frac{11}{11}$, or $f\frac{3}{11} = 1$. Therefore, when $x = \frac{3}{11}$, the value of fx is 1.

Q6**35**

The correct answer is 35. It's given that the perimeter of an isosceles triangle is 83 inches and that each of the two congruent sides has a length of 24 inches. The perimeter of a triangle is the sum of the lengths of its three sides. The equation $24 + 24 + x = 83$ can be used to represent this situation, where x is the length, in inches, of the third side. Combining like terms on the left-hand side of this equation yields $48 + x = 83$. Subtracting 48 from both sides of this equation yields $x = 35$. Therefore, the length, in inches, of the third side is 35.

Q7**A**

A. -12, 0

Choice A is correct. The x -intercept of a graph is the point where the graph intersects the x -axis. The graph of function f , where $y = fx$, intersects the x -axis at -12, 0. Therefore, the x -intercept of the graph of f is -12, 0.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**B****B.** 40

Choice B is correct. It's given that the shop's inventory starts with 4,500 paper cups and that the manager estimates that 70 of these paper cups are used each day. Let x represent the number of days in which the estimated supply of paper cups will reach 1,700. The equation $4,500 - 70x = 1,700$ represents this situation. Subtracting 4,500 from both sides of this equation yields $-70x = -2,800$. Dividing both sides of this equation by -70 yields $x = 40$. Therefore, based on this estimate, the supply of paper cups will reach 1,700 in 40 days.

Choice A is incorrect. After 20 days, the estimated supply of paper cups would be $4,500 - 70(20)$, or 3,100 cups, not 1,700 cups.

Choice C is incorrect. After 60 days, the estimated supply of paper cups would be $4,500 - 70(60)$, or 300 cups, not 1,700 cups.

Choice D is incorrect. After 80 days, the estimated supply of paper cups would be $4,500 - 70(80)$, or -1,100 cups, which isn't possible.

Q9**B****B.** 37

Choice B is correct. It's given that the function f is defined by $fx = 4x - 3$. Substituting 10 for x in the given function yields $f10 = 4(10) - 3$, which is equivalent to $f10 = 40 - 3$, or $f10 = 37$. Therefore, the value of $f10$ is 37.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f10$ for the function $fx = 4x$, not $fx = 4x - 3$.

Choice D is incorrect. This is the value of $f10$ for the function $fx = 4x + 3$, not $fx = 4x - 3$.

Q10

A

A. 14

Choice A is correct. Subtracting 180 from both sides of the given equation yields $5p = 70$. Dividing both sides of this equation by 5 yields $p = 14$. Therefore, the value of p that satisfies the equation $5p + 180 = 250$ is 14.

Choice B is incorrect. This value of p satisfies the equation $5p + 180 = 505$.

Choice C is incorrect. This value of p satisfies the equation $5p + 180 = 610$.

Choice D is incorrect. This value of p satisfies the equation $5p + 180 = 1,430$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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